Dr. Babasaheb Ambedkar Technological University, Lonere

B.Tech. in Chemical Engineering

Semester Examination: Nov 2017

Subject: Introduction to Material Science (CH 506)

Class/Semester: Third Year Chemical /V

Time: 3.00 Hrs **Date: 29/11/2017** **Marks:70**

Instructions to students

Question 1 is compulsory.

*Solve **five** Questions from remaining six questions*

Q.1 Do as directed **(10 M)**

1) Which type of composite materials the sandwich panels are?

a. Structural **b.** Fiber reinforced

c. Particle reinforced **d.** None of the above

2) What is meant by resilience in stress strain curve?

a. Area in the Plastic region **b.** Area in the elastic region

c. Area in Elastic and Plastic region **d.** None of the above

3) Draw the plane for (-2 0 -1)

4) .....+.....= Crystal Structure

5) Match the following group 1 items (crystal structures) with group 2 items (average number of atoms per unit cell) and select the correct option

1. Simple cubic crystal structure	A. 2
2. Body centered crystal structure	B. 6
3. Face centered crystal structure	C. 1
4. Hexagonal close packed structure	D. 4

a. 1 – A, 2 – C, 3 – D, 4 – B **b.** 1 – C, 2 – A, 3 – D, 4 – B

c. 1 – B, 2 – A, 3 – C, 4 – D **d.** 1 – D, 2 – A, 3 – B, 4 – C

6) In which type of point defect, positive and negative ions are missing from the crystal?

a. Vacancy defect **b.** Interstitial defect

c. Schottky defect **d.** Substitutional defect

7) ..... normally occurs after a lengthy period of repeated stress or strain cycling

8) Time dependent elastic behaviour is known as .....

9) Give the classification of composite material

10) Most crystalline solids are made up of millions of tiny single crystals called.....

Q.2. Attempt any two of the following:

a) Explain diamond structure in details ? **(6 M)**

b) **Write a short note on :** **(6 M)**

1) Grain boundaries 2) Tilt boundaries 3) Twin boundaries

c) Draw the plane and directions of FCC structures (321), (102), (111) **(6 M)**

Q.3. Explain

a) Theory of dislocation **(6 M)**

b) Rotating crystal method **(3 M)**

c) Crack initiation and propagation **(3 M)**

Q.4. Attempt the following

a) Explain true stress and strain **(4 M)**

b) Explain the example of Large Particle Composite **(4 M)**

c) Filament winding process **(4 M)**

Q.5. Explain the following

a) Polymer matrix composites **(6 M)**

b) Cyclic stresses in fatigue **(4 M)**

c) Strain Energy **(2 M)**

Q.6. Explain

a) Stress strain behaviour in ductile and brittle material **(4 M)**

b) What is creep? Explain its behaviour? **(4 M)**

c) A reinforced concrete column 500 mm * 500 mm in section is reinforced with 4 steel bars of 25 mm diameter, one in each corner . the column is carrying a load of 1000 KN. Find the stresses in the concrete and steel bars. Take E for steel = 210 GPa and E for concrete = 14 GPa. **(4 M)**

Q.7. Explain

a) One component phase diagram **(6 M)**

b) Homogeneous Nucleation **(6 M)**

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